

Time-Varying Break-Even Growth Rates for the Swiss Labour Market:

Line Manager Brief

June 1, 2026

1 Executive Summary

- Benchmark result: in the direct $Y \rightarrow U$ specification, the implied break-even GDP growth rate is about 1.9% annualized on average and about 2.1% annualized in the latest quarter, 2025 Q4.
- Time variation is present but modest under the preferred calibration with $\sigma_\eta = 0.05 \sigma_{OLS}$: the direct $Y \rightarrow U$ break-even series has a standard deviation of about 0.2 percentage points annualized.
- Additional model results: the $Y \rightarrow E$ specification implies a negative break-even GDP growth rate on average, while the $E \rightarrow U$ specification implies break-even employment growth of about 0.9% annualized on average and 1.2% annualized in 2025 Q4.
- Internal consistency check: chaining $Y \rightarrow E$ and $E \rightarrow U$ gives an implied $Y \rightarrow U$ break-even rate close to the direct estimate in levels, but more volatile over time.
- Practical read: the direct $Y \rightarrow U$ specification should remain the main benchmark; the employment-based specifications are useful companion views and interpretation aids.

Model	Break-even threshold	Scenario 1: $Y = 0\% / E = 0\%$	Scenario 2: $Y = 2\% / E = 1\%$
$Y \rightarrow U$	2.1	0.1	0.0
$Y \rightarrow E$	-1.2	0.4	1.1
$E \rightarrow U$	1.2	0.1	0.0

Table 1: Latest model readings for 2025 Q4. Growth rates are annualized.

2 Question and Motivation

- Objective: estimate the pace of Swiss real GDP growth needed to stabilize labour-market conditions, and allow that threshold to vary over time.
- Policy relevance: this is a compact way to summarize how labour-market resilience may be changing even if output growth, unemployment, and employment all remain noisy in the short run.
- Strategy: start from transparent reduced-form relationships, then let only the intercept vary over time in a local-level state-space model.

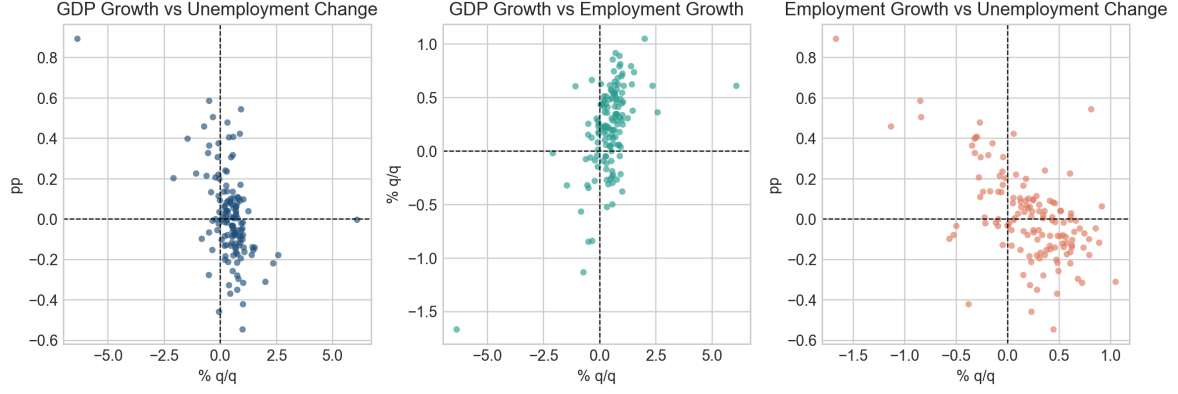


Figure 1: Motivation from the raw reduced-form relationships among GDP growth, employment growth, and unemployment changes.

3 Time-Invariant Evidence

- Baseline static reduced-form equation:

$$U_t = \alpha + \beta_0 Y_t + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \varepsilon_t.$$

- Estimated benchmark equation:

$$U_t = 0.123 - 0.108 Y_t - 0.092 Y_{t-1} - 0.070 Y_{t-2} + \varepsilon_t,$$

with $R^2 \approx 0.48$.

- Interpreting “break-even” as zero unemployment change and assuming constant GDP growth across current and lagged terms implies:

$$Y_t^* = -\frac{\alpha}{\beta_0 + \beta_1 + \beta_2}.$$

- Static benchmark: $Y_t^* \approx 0.455\%$ q/q, or about 1.83% annualized.
- ARDL robustness check:

$$U_t = \alpha + \phi U_{t-1} + \beta_0 Y_t + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \varepsilon_t.$$

- Estimated ARDL equation:

$$U_t = 0.054 + 0.714 U_{t-1} - 0.105 Y_t - 0.015 Y_{t-1} - 0.007 Y_{t-2} + \varepsilon_t,$$

with $R^2 \approx 0.80$ and much better residual behavior.

- The implied ARDL break-even GDP growth is close to the simpler benchmark: about 0.431% q/q, or 1.74% annualized.

4 Main Time-Varying Results

- We use the same local-level state-space routine across the three reduced-form relationships, but only illustrate the mechanics in detail for the benchmark $Y \rightarrow U$ specification.

- Preferred benchmark specification:

$$U_t = c_t + \beta_0 Y_t + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \gamma_2 D_{2020Q2,t} + \gamma_3 D_{2020Q3,t} + \varepsilon_t,$$

with

$$c_t = c_{t-1} + \eta_t.$$

- Estimation is done in `statsmodels` with a local-level model; the state innovation standard deviation is calibrated to $\sigma_\eta = 0.05 \sigma_{OLS}$.
- The implied break-even GDP growth rate in the benchmark model is:

$$Y_t^* = -\frac{c_t}{\beta_0 + \beta_1 + \beta_2}.$$

- Main findings: average break-even GDP growth is 1.86% annualized, the latest reading in 2025 Q4 is 2.09% annualized, and the annualized standard deviation is only about 0.20 percentage points.
- Scenario interpretation: under GDP stagnation, the model implies unemployment rises by about 0.12 percentage points on average; under GDP growth of 2% annualized, the implied unemployment change is essentially zero on average and 0.006 percentage points in 2025 Q4.
- Subsection 1, $Y \rightarrow E$:

$$E_t = c_t + \beta_0 Y_t + \beta_1 Y_{t-1} + \beta_2 Y_{t-2} + \gamma_2 D_{2020Q2,t} + \gamma_3 D_{2020Q3,t} + \varepsilon_t.$$

This maps GDP growth into employment growth rather than unemployment changes. It implies an average break-even GDP growth rate of -0.76% annualized and a latest reading of -1.18% annualized in 2025 Q4; under GDP growth of 2% annualized, implied employment growth is about 0.98% annualized on average and 1.14% annualized in 2025 Q4.

- Subsection 2, $E \rightarrow U$:

$$U_t = c_t + \theta_0 E_t + \theta_1 E_{t-1} + \theta_2 E_{t-2} + \gamma_2 D_{2020Q2,t} + \gamma_3 D_{2020Q3,t} + \varepsilon_t.$$

This yields a break-even employment-growth concept,

$$E_t^* = -\frac{c_t}{\theta_0 + \theta_1 + \theta_2}.$$

Average break-even employment growth is 0.91% annualized and the latest reading is 1.16% annualized in 2025 Q4; with employment growth of 1% annualized, implied unemployment change is close to zero on average and 0.017 percentage points in 2025 Q4.

- Bottom line across the three specifications: the direct $Y \rightarrow U$ model remains the most interpretable headline measure, while $Y \rightarrow E$ and $E \rightarrow U$ add complementary information on employment dynamics and labour-market transmission.

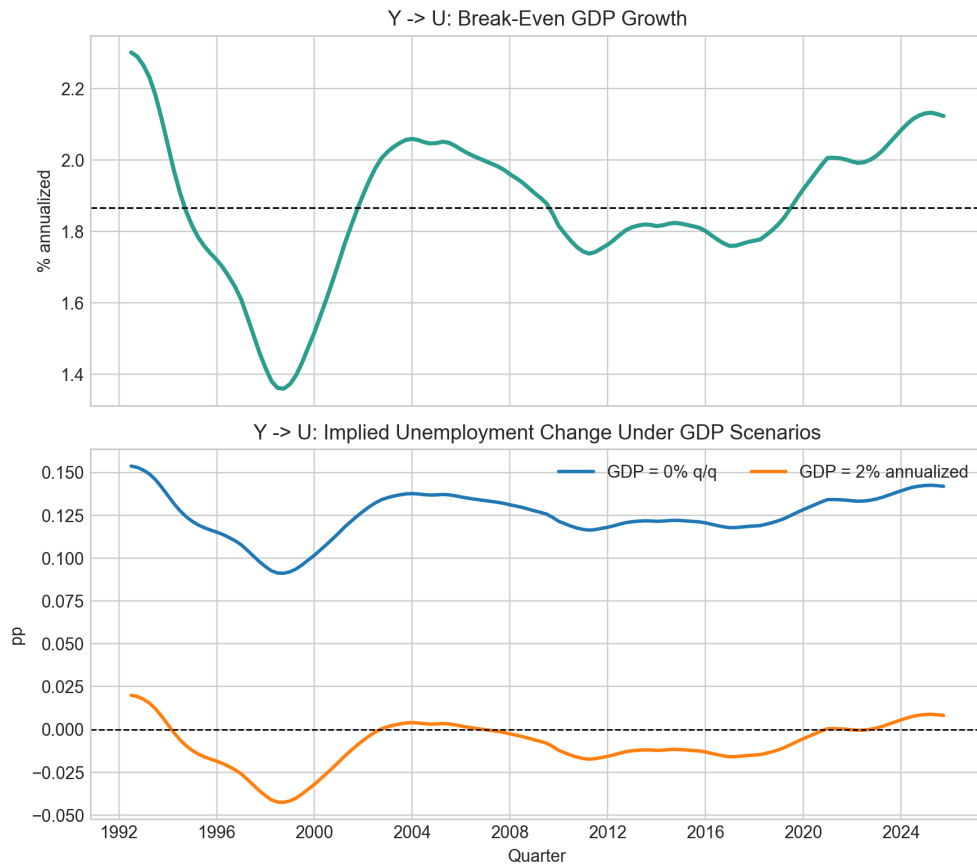


Figure 2: Preferred benchmark: time-varying $Y \rightarrow U$ specification. Top: implied break-even GDP growth. Bottom: implied unemployment change under two GDP scenarios.

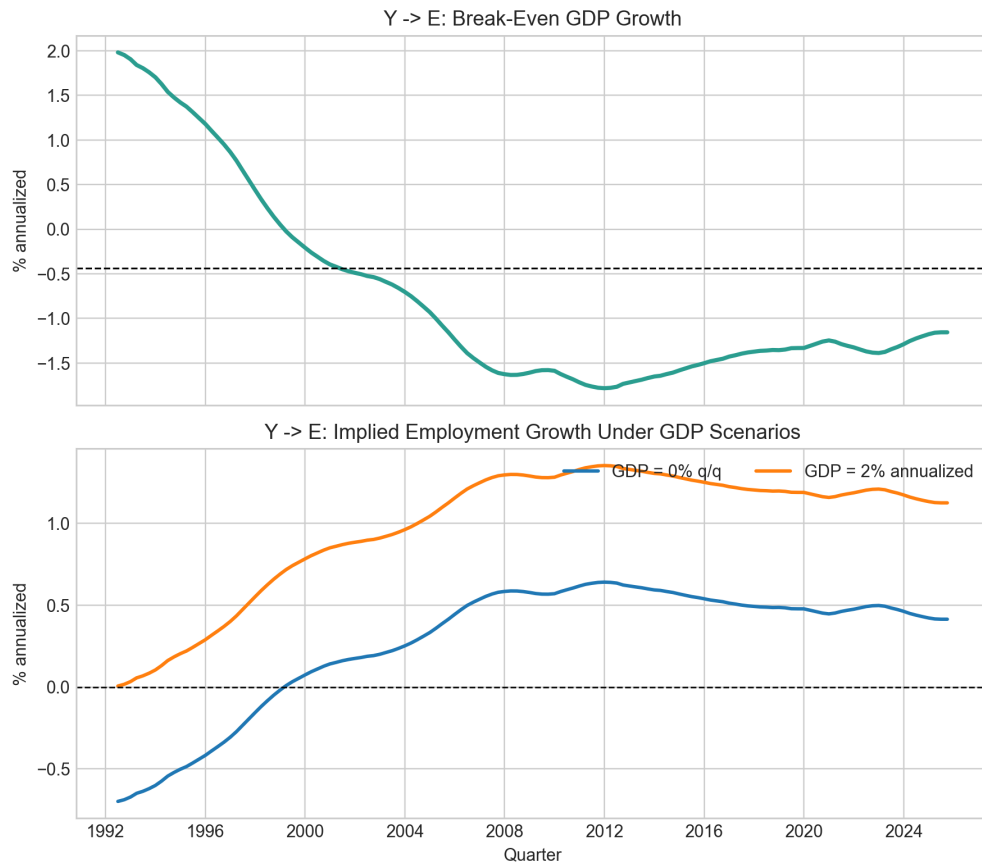


Figure 3: Time-varying $Y \rightarrow E$ specification. Top: implied break-even GDP growth for employment. Bottom: implied employment growth under two GDP scenarios.

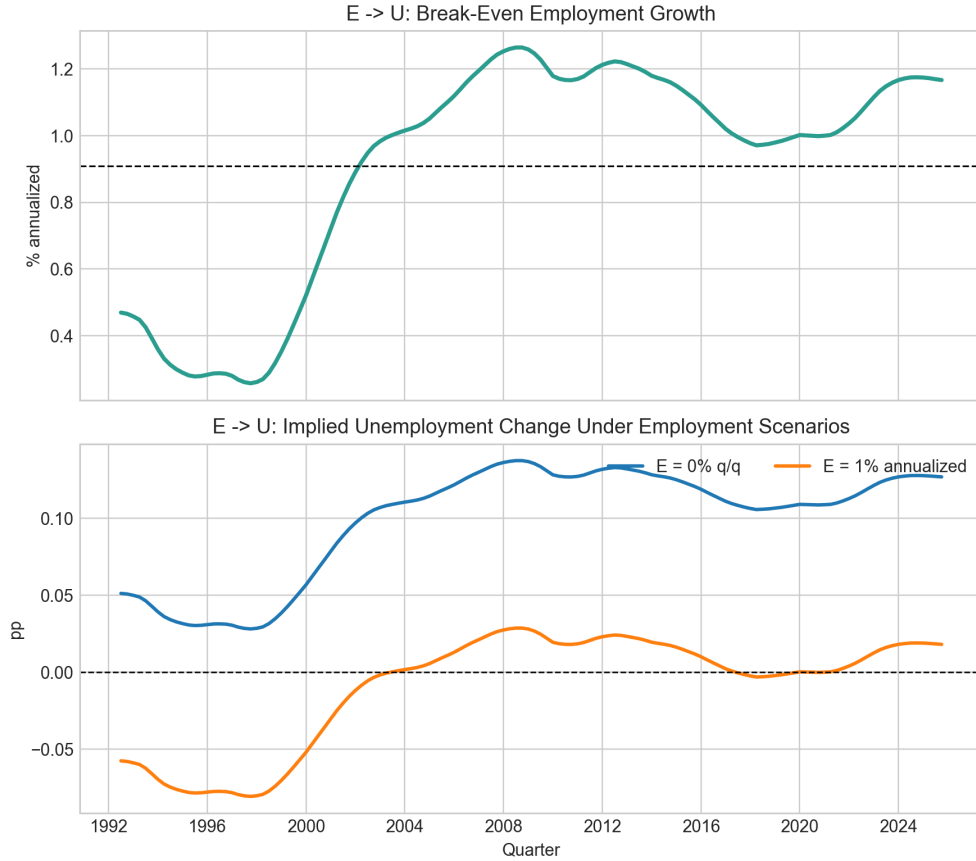


Figure 4: Time-varying $E \rightarrow U$ specification. Top: implied break-even employment growth. Bottom: implied unemployment change under two employment-growth scenarios.

5 Direct Versus Chained $Y \rightarrow U$

- A useful internal check is to combine the two employment-based steps: first $Y \rightarrow E$, then $E \rightarrow U$.
- In the static case, the implied chained break-even GDP growth rate is 1.80% annualized versus 1.86% annualized in the direct $Y \rightarrow U$ regression.
- In the time-varying case, the mean chained break-even rate is 1.79% annualized versus 1.86% annualized for the direct estimate.
- The main difference is volatility: the direct annualized series has a standard deviation of about 0.20 percentage points, while the chained series has a standard deviation of about 0.43 percentage points.
- Interpretation: the chained route broadly validates the level of the direct estimate, but it amplifies time variation and is therefore best read as a consistency exercise rather than the headline measure.

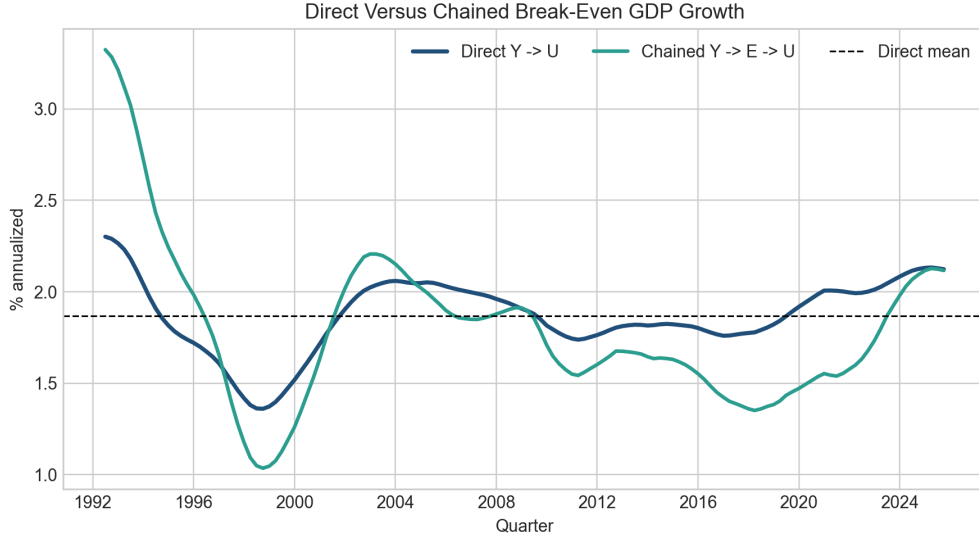


Figure 5: Direct versus chained break-even GDP growth.

6 A Diagnostic Tool: Labour-Implied GDP Growth

- As an additional diagnostic, we invert the labour-market equations and ask what GDP growth path would be most consistent with the observed unemployment path and the observed employment path.
- Concretely, we take the estimated time-varying labour-market equations as given, treat U or E as observed, treat Y as latent, and impose a separate AR(2) law of motion for GDP growth.
- The GDP AR(2) is estimated with the same two pandemic pulse dummies used elsewhere, but the inverse filter itself does not force those dummies into the latent GDP path.
- This is not an alternative measure of official GDP. It is a model-based consistency tool: if the labour-market equations are informative, the implied GDP lines should co-move with actual GDP.
- The results are encouraging but not exact. The unemployment-implied GDP series has a correlation of about 0.45 with actual GDP growth, while the employment-implied GDP series has a correlation of about 0.49.
- In the latest quarter, 2025 Q4, actual GDP growth is about 0.6% annualized, compared with 1.7% annualized for the unemployment-implied series and 1.0% annualized for the employment-implied series.
- Practical interpretation: this tool is useful as a triangulation device. When the labour-implied GDP lines move well away from measured GDP, that flags either unusual labour-market adjustment or limits of the reduced-form mapping.

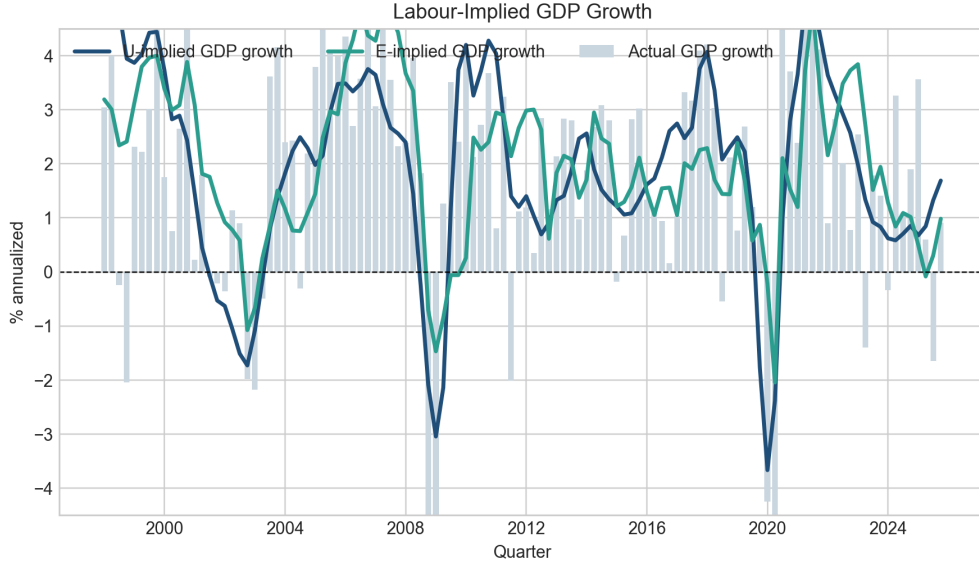


Figure 6: Actual GDP growth and the GDP growth rates implied by the unemployment and employment equations.

7 Conclusion

- The cleanest headline result remains the direct $Y \rightarrow U$ specification.
- On that benchmark, Swiss break-even GDP growth is close to 1.9% annualized on average and just above 2.0% annualized in the latest quarter.
- The preferred calibration implies meaningful but not excessive time variation.
- The employment-based specifications are useful corroborating evidence: they broadly support the level of the direct benchmark when chained together, but they are less stable as standalone break-even measures.
- Recommended communication line: “A reduced-form time-varying benchmark suggests that Swiss GDP growth of around 2% annualized is currently needed to keep unemployment broadly stable.”

A Data Construction Summary

- Employment: a blended quarterly employment series is constructed by averaging q/q growth from the seasonally adjusted household and firm surveys, cumulating the average growth rate into an index, and scaling levels to match the household survey average in 2019.
- Household survey: the raw household-employment series is seasonally adjusted back to the 1970s using a quarterly STL-based procedure and then chained to the official seasonally adjusted BFS series from 2010 onward.
- Unemployment: the raw monthly seasonally adjusted series is aggregated to quarterly averages.
- GDP: the memo uses seasonally, calendar-, and sport-adjusted real GDP from SECO and its q/q growth rate.

- Final analysis variables: E is q/q growth of the constructed employment series, U is the quarterly change in the unemployment rate, and Y is q/q GDP growth.

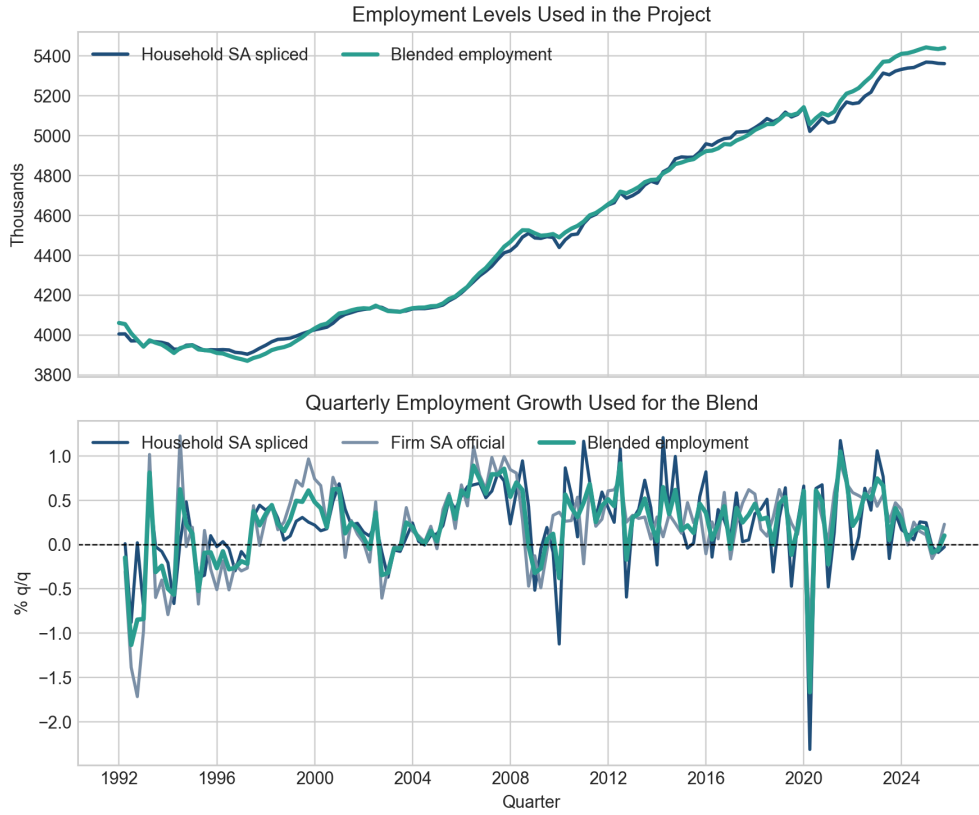


Figure 7: Data construction: the project blends household and firm employment information while preserving the household-level anchor.

B Data Appendix: Key Choices from Notebook 02

- The project uses both the BFS household and firm employment concepts rather than choosing only one source.
- The long household series is back-extended via internal seasonal adjustment, benchmarked to the official overlap.
- The final blended employment measure is intended to preserve household-level scaling while smoothing quarter-to-quarter survey-specific noise.
- Quarterly aggregation choices matter: unemployment is quarter-average, while GDP and employment are used in quarterly growth-rate form.

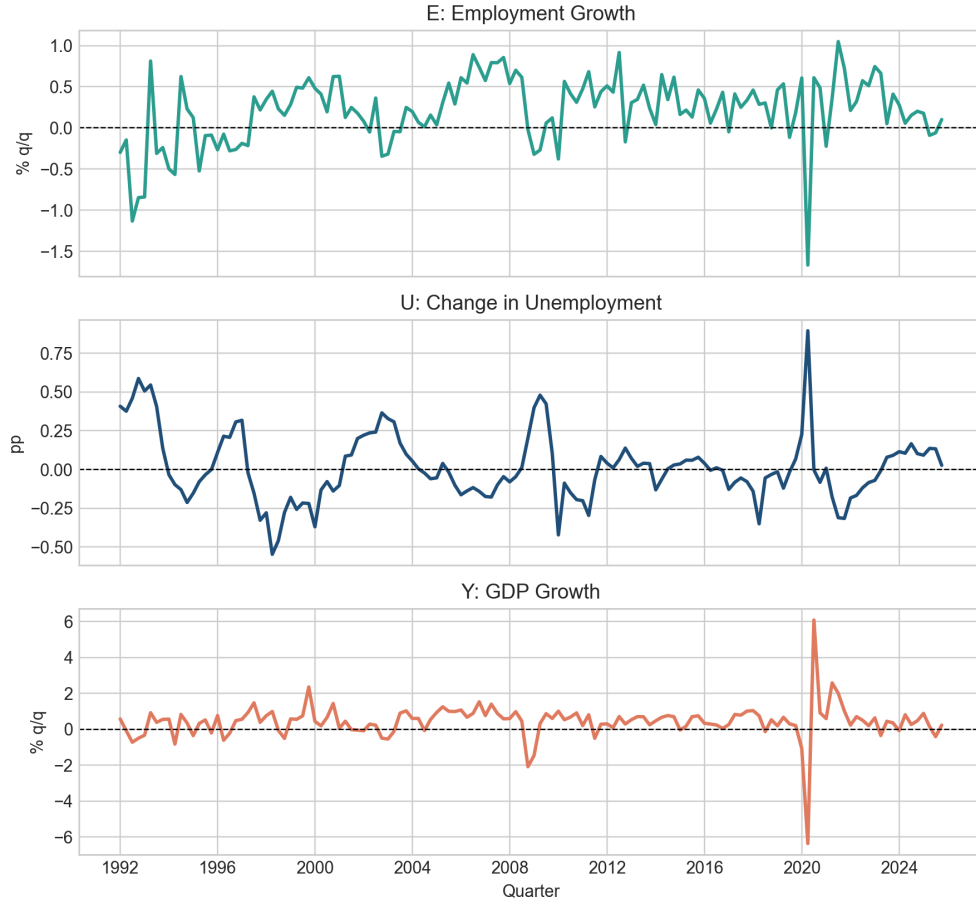


Figure 8: Final quarterly analysis dataset used in estimation: employment growth (E), change in unemployment (U), and GDP growth (Y).

C Empirical Caveats

- These are reduced-form relationships, not structural labour-market models.
- Results depend on the choice of calibration for the state variance; the memo uses $\sigma_\eta = 0.05 \sigma_{OLS}$ as a deliberately smooth benchmark.
- The COVID pulse dummies improve robustness, but unusual episodes can still influence the estimated latent intercept.
- The $Y \rightarrow E$ specification gives a different break-even interpretation from the direct $Y \rightarrow U$ benchmark and should not be over-interpreted on its own.
- The chained $Y \rightarrow E \rightarrow U$ comparison is an internal consistency check, not an independent identification strategy.